

BCA-5003: COMPUTER NETWORK

UNIT II — COMPLETE DETAILED NOTES + TOPIC-WISE PYQ ANSWER BOOK

Transmission Media • Satellite Communication • Transmission Impairments • Data-Rate Limits

Exact Unit-I format: Index → detailed concept notes → diagrams/graphs → topic-wise PYQs with years → long model answers → revision sheets

Prepared for long university answers, diagram practice, formula numericals, and 10-CGPA-oriented preparation.

Index

No.	Topic	PYQ focus in supplied Unit-II material
1	Unit-II roadmap and answer-writing method	—
2	Transmission media: meaning and classification	2018; 2024–25
3	Guided media: UTP, STP, coaxial cable, optical fibre	2014, 2018, 2024–25, 2025–26
4	Unguided media: radio, RF propagation, microwave, infrared	2017, 2018
5	Satellite communication and satellite networks	2022–23, 2023–24
6	Transmission impairment: attenuation, distortion, noise	2021
7	Noiseless and noisy channels	2017, 2021, 2022–23
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How to use this Unit

1. Read the concept notes first and underline definitions, formula variables, advantages, limitations, and applications.
2. Practice every labelled diagram shown in this PDF. In a theory paper, a clean diagram can substantially improve the presentation of a 5–10 mark answer.
3. After each topic, learn the PYQ answer in the same structure: introduction → diagram → headings → explanation → conclusion.
4. For Nyquist/Shannon questions, write formula, units, substitution, calculation, and final answer in a box.

1. Unit-II Roadmap

Unit II explains how signals physically travel from one network device to another and what limits their speed and quality. It is the foundation for understanding the OSI Physical Layer, LAN cabling, Wi-Fi, satellite links, signal quality, and data-rate numericals.

Study order for maximum marks

1. First learn guided and unguided media classification.
2. Then master each cable's structure, working, advantages, disadvantages, and applications.
3. Learn radio propagation, microwave line-of-sight communication, and satellite terms.
4. Understand the three impairments: attenuation, distortion, and noise.
5. Memorize Nyquist and Shannon formulas and practise calculations.
6. Finally, write all PYQs from memory under timed conditions.

High-score answer template

For an “Explain transmission media” answer, do not merely list names. Start with definition, draw the classification diagram, explain every medium under separate headings, include structure/working, then end with an appropriate comparison or conclusion.

2. Transmission Media: Meaning and Classification

Transmission media are the channels through which data signals travel from a sender to a receiver. The signal may be electrical in copper cable, optical in fibre cable, or electromagnetic in wireless/free-space communication. The choice of medium affects data capacity, transmission distance, attenuation, interference, cost, installation, security, and mobility.

Classification diagram

TRANSMISSION MEDIA
■■■ Guided / bounded / wired media – signal follows a physical path
■ ■■■ Twisted-pair cable: UTP and STP
■ ■■■ Coaxial cable
■ ■■■ Optical fibre cable
■■■ Unguided / unbounded / wireless media – signal travels through air/free space
■■■ Radio waves
■■■ Microwaves
■■■ Infrared
■■■ Satellite communication

Guided media provide a controlled physical route. They are commonly preferred for fixed installations such as offices, computer laboratories, and data centres. Unguided media support mobility and places where cabling is difficult or expensive, but their performance can be affected by obstacles, weather, interference, spectrum availability, and security threats.

Feature	Guided media	Unguided media
Path	Signal is guided through physical cable/fibre.	Signal propagates through air, vacuum, or free space.
Common examples	UTP, STP, coaxial, optical fibre.	Radio, Wi-Fi, microwave, infrared, satellite.
Mobility	Low; cable connection is needed.	High; endpoints can move within coverage.
Interference	Copper can face EMI; fibre is highly immune.	More affected by obstacles, weather, interference, and spectrum congestion.
Security	Physical access to the cable is generally needed.	Signals can be intercepted more easily unless encryption is used.

Key definition for exam

Transmission media are the physical or wireless communication channels that carry data signals between sender and receiver. They are classified into guided media, in which signals follow cables, and unguided media, in which electromagnetic waves travel through free space.

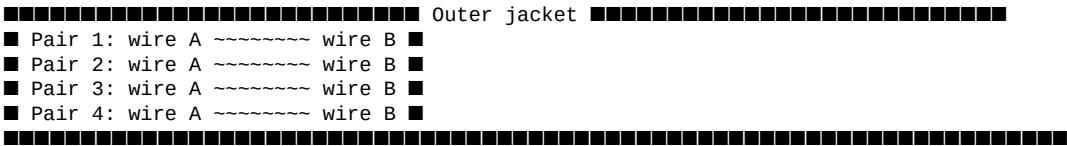
3. Guided Transmission Media

Guided media use cables or fibres as a physical path. Copper cables carry electrical signals. Optical fibre carries modulated light pulses. The major guided media are twisted-pair cable, coaxial cable, and optical fibre.

3.1 Twisted-Pair Cable

A twisted-pair cable contains two insulated copper conductors twisted around each other. Twisting reduces noise pickup and crosstalk because external electromagnetic interference tends to affect both conductors similarly; differential signalling can reject much of this common interference. Standard Ethernet LAN cable commonly contains four twisted pairs enclosed inside one outer jacket.

Diagram: four-pair Ethernet cable



UTP — Unshielded Twisted Pair

UTP consists of insulated twisted copper pairs with no metallic shield. It is lightweight, flexible, inexpensive, easy to terminate, and widely used in Ethernet LANs and telephone cabling. Modern categories such as Cat5e, Cat6, Cat6A, and beyond are designed for different frequency/performance requirements.

- Advantages:** low cost; flexible; easy installation; readily available; suitable for voice and LAN data.
- Limitations:** copper is vulnerable to electromagnetic interference, crosstalk, and attenuation; cable-distance limits apply.

STP — Shielded Twisted Pair

STP includes twisted pairs plus metallic foil/braid shielding. The shield helps reduce electromagnetic interference and crosstalk in noisy electrical environments, such as industrial sites or locations near heavy machinery. Shielding should be properly grounded to work effectively.

- Advantages:** better EMI protection and potentially better performance in electrically noisy locations.
- Limitations:** thicker, heavier, less flexible, more expensive, and more difficult to install than UTP.

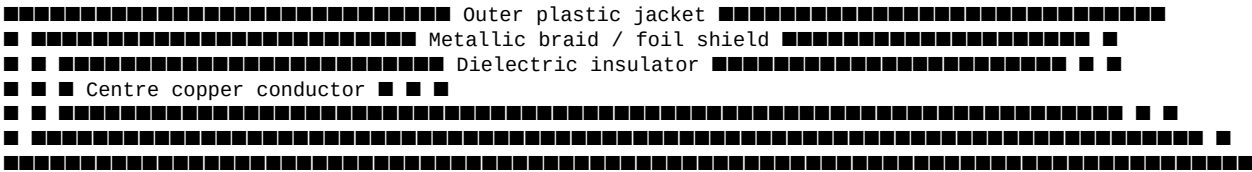
UTP versus STP

Basis	UTP	STP
Shield	No metallic shield.	Foil/braid metallic shield included.
Cost	Lower.	Higher.
Flexibility	More flexible and easier to install.	Less flexible and installation is more demanding.
EMI immunity	Moderate.	Higher when correctly grounded.
Typical use	Homes, offices, labs, standard LANs.	Factories, medical/industrial or high-EMI environments.

3.2 Coaxial Cable

Coaxial cable has two conductors sharing a common axis. The central conductor carries the signal; a dielectric insulator surrounds it; then an outer conducting shield surrounds the dielectric; a protective jacket covers the cable. The shield protects the inner signal from external interference and confines the electromagnetic field.

Diagram: coaxial cable cross-section

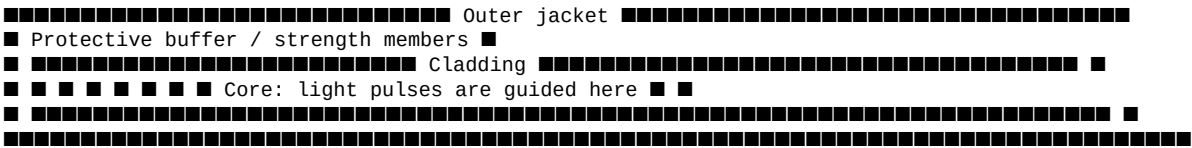


Coaxial cable generally provides better noise resistance and supports higher-frequency signalling than ordinary twisted pair, but it is thicker and more expensive. It has been used for cable television, CCTV, RF distribution, broadband systems, and older Ethernet networks.

3.3 Optical Fibre Cable

Optical fibre is a strand of very pure glass or plastic that carries information as pulses of light. It offers exceptionally high bandwidth, low attenuation, low bit-error probability, electrical isolation, and immunity to electromagnetic interference. It is widely used in telecom backbone networks, data centres, fibre broadband, and long-distance communication.

Diagram: optical fibre construction



Working principle: the core has a slightly higher refractive index than the cladding. When light in the core strikes the core–cladding boundary at an angle greater than the critical angle, it undergoes **total internal reflection** and remains guided within the core. This is the correct scientific explanation of light guidance in optical fibre.

Types of fibre

Type	Description	Key use
Single-mode fibre	Small core; essentially one light mode/path. Very low dispersion and very high long-distance capacity.	Long-haul telecom and Internet backbone.
Multimode step-index fibre	Larger core; many rays/modes take different paths. Abrupt refractive-index change at cladding boundary.	Short distance/older installations.
Multimode graded-index fibre	Refractive index gradually decreases from core centre outward, reducing modal dispersion.	Campus, building, and data-centre links.

Advantages of optical fibre

- Very high bandwidth and support for very high data rates.
- Low attenuation, so longer distances are possible with fewer repeaters/regenerators.
- Immunity to electromagnetic interference, radio-frequency interference, and crosstalk.
- Better physical security because fibre does not radiate electromagnetic signals like copper; however, no medium is absolutely impossible to tap.
- Small diameter, low weight, electrical isolation, and no spark/fire risk from signal current.
- High reliability and potentially low long-term operational cost for large-capacity links.

Limitations of optical fibre

- Higher initial equipment, connector, and installation cost than basic copper cabling.
- Fibre is mechanically more delicate and requires skilled splicing/termination and specialised test equipment.
- It cannot deliver electrical power to endpoint devices as copper Ethernet can with Power over Ethernet.

4. Unguided Transmission Media

Unguided media transmit electromagnetic signals through air, vacuum, or free space without a physical cable between sender and receiver. They are used when mobility, broad coverage, rapid deployment, or difficult terrain makes wired infrastructure less suitable.

4.1 Radio Waves and RF Propagation

Radio waves are electromagnetic waves used for broadcasting, mobile communication, Wi-Fi, Bluetooth, wireless LANs, and many other services. Depending on frequency and environment, radio signals may propagate as ground waves, sky waves, or line-of-sight waves.

Propagation	Diagram / behaviour	Example / note
Ground wave	Signal follows Earth's curvature near the surface.	Lower-frequency radio; AM broadcasting is the classic textbook example.
Sky wave / ionospheric	Signal returns toward Earth after interaction with ionosphere.	Useful for long-distance HF radio; varies with time and atmospheric conditions.
Line of sight (LOS)	Transmitter and receiver need a direct unobstructed/near-unobstructed path.	FM, terrestrial microwave, many wireless bridges, satellite ground links.

RF propagation sketches

4.2 Microwave Communication

Diagram: terrestrial microwave link

4.3 Infrared

4.4 Satellite

5. Satellite Communication

Main diagram: satellite communication system

Important terminology

Satellite link concept

BCA-5003 Computer Network — Unit II Detailed Notes + Topic-Wise PYQ Answers

Applications

- Television and radio broadcasting.
- Telephone, Internet, and enterprise connectivity in remote regions.
- Navigation, weather monitoring, disaster recovery, defence, maritime, and aviation communication.
- Wide-area broadcast/multicast distribution and emergency backup networks.

Advantages

- Very large geographical coverage, including remote, mountainous, oceanic, and disaster-affected locations.
- A single satellite beam can serve/broadcast to a large coverage area.
- Useful where terrestrial fibre, cable, or microwave tower infrastructure is unavailable.

Limitations

- High cost of satellites, ground equipment, spectrum coordination, and operation.
- GEO links have significant propagation delay because of the long signal path to space and back.
- Weather, especially rain at high-frequency bands, can weaken signals (rain fade).
- Capacity and spectrum must be carefully managed.

6. Transmission Impairment

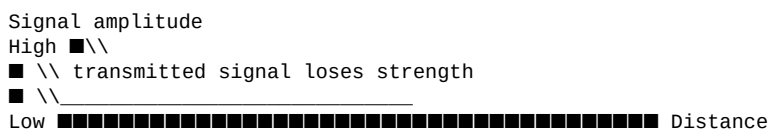
Transmission impairment means deterioration or alteration of a signal as it travels through a communication medium. The signal received at the destination may not have exactly the same strength, shape, timing, or information content as the transmitted signal. For digital data, impairment can cause bit errors; for analog signals, it reduces quality.

The three major types are attenuation, distortion, and noise.

6.1 Attenuation

Attenuation is the gradual reduction in signal strength as distance increases. It is commonly measured in decibels (dB). If a signal becomes too weak compared with background noise, the receiver may misinterpret bits or lose the signal. Digital repeaters regenerate signals; amplifiers increase signal power but can also amplify noise.

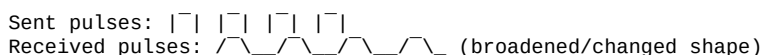
Graph: attenuation with distance



6.2 Distortion

Distortion occurs when different frequency components of a composite signal experience different attenuation or propagation delay. This changes the waveform shape. In digital systems, pulse spreading can cause intersymbol interference, where one symbol overlaps another and the receiver makes incorrect decisions.

Graph: pulse distortion



6.3 Noise

Noise is unwanted electrical or electromagnetic energy that becomes mixed with the desired signal. It is random and unpredictable. It cannot be removed completely, but good design can reduce its impact. Noise lowers signal-to-noise ratio (SNR), increases bit errors, and limits maximum reliable data rate.

Graph: desired signal with noise

Amplitude
Desired: ■■■■ ■■■■ ■■■■
With noise: ~~~/_~ ~~~/_~ ~~~/_~ (random unwanted variation)
Time →

Noise type	Description
Thermal noise	Random electron movement due to temperature; unavoidable in electronic components.
Induced noise / EMI	Interference from power lines, motors, switching equipment, and nearby electromagnetic sources.
Crosstalk	Unwanted coupling in which signals from one cable/channel affect another.
Impulse noise	Short, high-amplitude spikes due to lightning, switching, faults, etc.; can create burst errors.

Signal-to-noise ratio

SNR compares desired signal power S with noise power N. Higher SNR usually means clearer reception and higher potential data rate. In decibels:

$$\text{SNR(dB)} = 10 \log_{10}(\text{S/N})$$

For Shannon's formula, S/N must be the ordinary linear ratio, not the dB value. If SNR is given in dB, convert using: $\text{S/N} = 10^{(\text{SNR(dB)}/10)}$.

7. Data-Rate Limits: Nyquist and Shannon

Data rate is the number of bits transmitted per second and is measured in bps. The maximum possible rate is not unlimited. It depends on available bandwidth, signal levels, and channel quality/noise. Two fundamental formulas describe theoretical limits: Nyquist for an ideal noiseless channel and Shannon for a noisy real channel.

Term	Meaning
Bandwidth (B)	Frequency range available in a channel; measured in hertz (Hz).
Bit rate	Number of data bits sent per second; measured in bps.
Baud rate	Number of symbols/signal elements transmitted per second. A symbol can carry one or more bits.
Signal levels (L)	Number of distinct signal states used to represent data.
SNR	Signal power divided by noise power.

7.1 Nyquist Bit-Rate Formula: Noiseless Channel

For an ideal channel without noise, Nyquist gives the maximum bit rate:

$$\text{Maximum bit rate} = 2B \log_2(L) \text{ bps}$$

Where B is bandwidth in hertz and L is the number of signal levels. Larger bandwidth permits higher bit rate. More levels can theoretically carry more bits per symbol, but in real channels many levels are harder to distinguish and reduce reliability.

Solved example 1

Given B = 3000 Hz and L = 2

$$\text{Bit rate} = 2 \times 3000 \times \log_2(2)$$

$$= 6000 \times 1$$

$$= \mathbf{6000 \text{ bps}}$$

Solved example 2

Given B = 3000 Hz and L = 8

$$\text{Bit rate} = 2 \times 3000 \times \log_2(8)$$

$$= 6000 \times 3$$

$$= \mathbf{18,000 \text{ bps}}$$

7.2 Shannon Capacity: Noisy Channel

Every practical communication channel has noise. Shannon's theorem gives the highest theoretical reliable data rate for a noisy channel:

Capacity = $B \log_2(1 + S/N)$ bps

Where B is bandwidth in Hz and S/N is the linear signal-to-noise power ratio. Unlike Nyquist, Shannon's limit does not contain the number of signal levels. It is a fundamental upper limit imposed by the channel's bandwidth and noise.

Solved example 3

Given B = 3000 Hz and S/N = 1000

Capacity = $3000 \times \log_2(1 + 1000)$

= $3000 \times \log_2(1001)$

≈ 3000×9.967

≈ **29,900 bps**

Solved example 4: SNR given in dB

Given B = 3000 Hz and SNR = 30 dB

First convert: $S/N = 10^{(30/10)} = 10^3 = 1000$

Then: Capacity = $3000 \times \log_2(1001) \approx \mathbf{29,900 \text{ bps}}$

Nyquist versus Shannon

Basis	Nyquist theorem	Shannon capacity
Channel	Ideal noiseless channel.	Practical noisy channel.
Formula	$2B \log_2 L$	$B \log_2(1 + S/N)$
Depends on	Bandwidth and number of signal levels.	Bandwidth and linear SNR.
Meaning	Theoretical rate from bandwidth and signalling levels.	Absolute upper reliable capacity of a noisy channel.
Practical observation	Increasing L increases theoretical rate but makes detection difficult.	No signalling scheme can reliably exceed Shannon capacity.

8. Topic-Wise PYQ Answer Bank

These answers follow the Unit-I pattern: the concept is explained above; then the exact/near-exact PYQ is shown with its year and a detailed model answer.

Q1. Mention the advantages of optical fibre cables.

Asked: 2025–26

Answer

Optical fibre is a guided communication medium that transmits data as pulses of light through a glass or plastic core. It is widely used for high-speed and long-distance communication because of its superior performance compared with copper cable.

- 1. High bandwidth:** Optical fibre can carry very large volumes of information at high speed. It is suitable for broadband Internet, telecommunications, high-definition video, cloud services, and data centres.
- 2. Low attenuation:** Light signals experience very low loss in fibre. Therefore, communication can be extended over long distances with fewer repeaters/regenerators than copper systems.
- 3. Immunity to electromagnetic interference:** Fibre does not carry electrical current. It is not affected by electromagnetic interference, radio-frequency interference, lightning-induced noise, or crosstalk like copper cables.
- 4. Better security:** Fibre does not radiate electromagnetic signals in the same way as copper. Unauthorised tapping is more difficult and normally requires physical access to the cable, making fibre relatively secure.
- 5. Small size and light weight:** Fibre cables are thin and light, so a large capacity can be installed in limited duct space.
- 6. Electrical isolation and safety:** Fibre is non-conductive. It does not create spark hazards or ground loops and is suitable in high-voltage or electrically noisy environments.
- 7. High reliability and low error rate:** Its EMI immunity and low attenuation support high-quality transmission with low bit error probability.
- 8. Lower long-term operating cost:** While initial deployment can be expensive, fewer repeaters, high capacity, and reliability can make fibre economical over time.

Conclusion: Optical fibre is the preferred medium for high-bandwidth, long-distance, secure, and interference-resistant communication.

Q2. Explain various transmission media in detail.

Asked: 2018; related indicators: 2014, 2024–25

Answer

Transmission media are the communication paths that carry signals from a sender to a receiver. They are classified as guided media and unguided media.

Guided media carry signals through a physical path. Twisted-pair cable uses insulated copper wires twisted together. UTP is unshielded, economical, flexible, and common in LANs. STP includes shielding and is better in electrically noisy environments. Coaxial cable has a central conductor, dielectric, metallic shield, and outer jacket; it offers good noise immunity and is used in cable TV, CCTV, and RF systems. Optical fibre carries light through a core and cladding by total internal reflection; it provides very high bandwidth, low attenuation, long distance, security, and immunity to EMI.

Unguided media transmit electromagnetic waves through free space. Radio waves are used for broadcasting and wireless networking. Microwaves are directional and usually line-of-sight; they are used in terrestrial point-to-point links and satellite communication. Infrared is short-range and generally does not penetrate walls. Satellite communication uses a satellite transponder to relay signals between distant Earth stations through uplink and downlink paths.

Conclusion: Guided media are appropriate for fixed, controlled installations; unguided media are suitable for mobility, wide coverage, and hard-to-cable regions. Medium selection depends on bandwidth, distance, cost, interference, security, and mobility requirements.

Q3. Write two types of guided media and two types of unguided media.

Asked: 2018

Answer

Guided media: Twisted-pair cable (UTP/STP) and coaxial cable. Optical fibre is another guided medium.

Unguided media: Radio waves and microwaves. Infrared and satellite communication are also unguided media.

Guided media use a physical cable to guide signals, whereas unguided media carry electromagnetic waves through air or free space.

Q4. What is satellite? Explain satellite communication with a proper diagram.

Asked: 2022–23; satellite networks: 2023–24

Answer

A communication satellite is an artificial satellite equipped with transponders that receives radio signals from an Earth station and retransmits them to another Earth station. It works as a relay/repeater in space and enables communication over very large geographical distances.

Diagram:

Earth station A → Uplink → Satellite transponder (receive → filter → frequency conversion → amplification → transmit) → Downlink → Earth station B.

Working: The sender Earth station sends an uplink signal to the satellite. The satellite transponder receives it, filters it, usually changes its frequency to separate receive and transmit channels, amplifies it, and sends it back to the coverage area as a downlink. A receiving Earth station receives and demodulates the downlink.

Terms: Uplink means Earth-to-satellite transmission; downlink means satellite-to-Earth transmission; a footprint is the satellite coverage area; a GEO satellite appears stationary because it orbits Earth at the same angular speed as Earth's rotation.

Applications: TV/radio broadcast, remote Internet, telephone, weather and navigation services, military, aviation, maritime communication, and disaster response.

Advantages: wide coverage, remote-area connectivity, broadcast capability, and deployment where terrestrial infrastructure is difficult.

Limitations: high cost, weather effects, and high propagation delay for GEO systems.

Conclusion: Satellite communication is essential for wide-area connectivity, especially where terrestrial links are unavailable or impractical.

Q5. What is transmission impairment? Explain noise in signal transmission.

Asked: 2021

Answer

Transmission impairment is deterioration of a signal during its travel through a communication medium. The signal received at the destination may not be identical to the signal transmitted. Digital signals may suffer bit errors; analog signals may lose quality.

The main impairments are attenuation, distortion, and noise. Attenuation is loss of signal strength with distance. Distortion is waveform change because components travel differently. Noise is an unwanted random signal added to the desired message signal.

Noise: Noise can be introduced by electronic components, power lines, motors, nearby cables, atmospheric events, and switching activity. It makes valid signal levels difficult to distinguish, thereby lowering SNR and increasing errors.

Types: thermal noise due to random electron motion; induced/EMI noise from electrical sources; crosstalk due to coupling between lines; and impulse noise due to sudden high-energy events such as lightning or switching.

Effects: hiss in radio, buzz in telephone audio, flicker in television, corrupted bits in digital data, and lower maximum reliable data rate.

Reduction methods: shielding, twisted pair, proper grounding, fibre optics, filtering, better cable routing, stronger signal design, and error detection/correction techniques.

Conclusion: Noise cannot be eliminated completely, but it can be controlled so that the receiver can recover data reliably.

Q6. Differentiate between noiseless and noisy channels.

Asked: 2017, 2021, 2022–23

Answer

Noiseless channel: An ideal theoretical channel in which random noise is assumed absent. Nyquist theorem is used: Maximum bit rate = $2B \log_2 L$. The rate depends on bandwidth B and signal levels L.

Noisy channel: A practical channel that contains noise. Shannon capacity is used: Capacity = $B \log_2(1 + S/N)$. The rate depends on bandwidth and signal-to-noise ratio.

Differences:

1. Noise: absent in ideal noiseless model; present in noisy channel.
2. Formula: Nyquist for noiseless; Shannon for noisy.
3. Factors: Nyquist uses B and L; Shannon uses B and S/N.
4. Reality: noiseless is an analytical assumption; all real channels have some noise.
5. Error probability: ideal noiseless model assumes no noise-caused error; noisy channels require robust signalling/coding for reliable communication.

Q7. What are data-rate limits? On which factors do they depend? Discuss calculation for noisy and noiseless channels.

Asked: 2015

Answer

Data-rate limit is the highest theoretical rate at which data can be transmitted reliably through a channel. It is measured in bits per second. The major factors are available bandwidth, number of signal levels, and channel quality/noise level.

For a noiseless channel: Nyquist formula applies: Maximum bit rate = $2B \log_2 L$ bps. Here B is bandwidth in Hz and L is number of distinct signal levels. Example: if B = 3000 Hz and L = 2, rate = $2 \times 3000 \times \log_2 2 = 6000$ bps.

For a noisy channel: Shannon formula applies: Capacity = $B \log_2(1 + S/N)$ bps. Here S/N is linear signal-to-noise ratio. Example: B = 3000 Hz, S/N = 1000. Capacity = $3000 \log_2(1001) \approx 29,900$ bps.

Conclusion: Nyquist explains ideal-channel limits based on bandwidth and levels, while Shannon defines the fundamental maximum reliable rate in a real noisy channel. Practical systems must remain within channel constraints.

Q8. Differentiate guided and unguided transmission media. Give one example of each.

Asked: 2017

Answer

Guided media carry signals through a physical path such as copper cable or optical fibre. Examples: UTP, coaxial cable, and optical fibre. They have controlled paths, require cable installation, generally offer more stable performance, and are physically harder to intercept.

Unguided media carry electromagnetic waves through air, vacuum, or free space. Examples: radio waves, microwave, infrared, and satellite. They provide mobility and broad coverage but can be affected by weather, obstacles, interference, and interception unless protected by security mechanisms.

One example: Optical fibre is guided media; microwave is unguided media.

9. Final Revision Sheet

Definitions to memorise

Term	One-line answer
Transmission medium	Physical or wireless channel that carries data signals between devices.
Guided media	Media in which signals travel through a physical cable/fibre path.
Unguided media	Media in which electromagnetic waves travel through air or free space.
Attenuation	Reduction of signal strength with distance.
Distortion	Change in waveform shape because components experience unequal delay/attenuation.
Noise	Unwanted random energy/signal added to the desired signal.
SNR	Ratio of desired signal power to noise power.
Uplink	Earth station to satellite communication.
Downlink	Satellite to Earth station communication.
Transponder	Satellite subsystem that receives and retransmits a processed radio signal.

Formula box

Nyquist: Maximum bit rate = $2B \log_2 L$ bps

Shannon: Capacity = $B \log_2(1 + S/N)$ bps

SNR in dB: $SNR(dB) = 10 \log_{10}(S/N)$

dB to linear: $S/N = 10^{(SNR(dB)/10)}$

Diagrams you must practise

- Classification of guided and unguided transmission media.
- UTP/STP twisted-pair cable diagram.
- Coaxial cable layered cross-section.
- Optical fibre: core, cladding, jacket, total internal reflection path.
- Ground wave, sky wave, and line-of-sight propagation.
- Terrestrial microwave tower/relay link.
- Satellite communication: Earth station, uplink, transponder, downlink, footprint.
- Attenuation graph and noise-on-signal waveform.

Most expected questions

1. Explain various guided and unguided transmission media in detail.
2. Explain optical fibre cable with structure, working, advantages, limitations, and applications.
3. Explain satellite communication with a neat diagram.
4. What is transmission impairment? Explain attenuation, distortion, and noise.
5. Differentiate noisy and noiseless channels. State Nyquist and Shannon formulas.
6. Solve numerical problems using Nyquist and Shannon capacity formulas.
7. Differentiate guided and unguided transmission media.

7-day Unit-II 10-CGPA plan

Day	Study target	What to write/practise
1	Media classification and UTP/STP	Classification and UTP/STP diagrams; PYQ Q2/Q3.
2	Coaxial and optical fibre	Cross-section and fibre working; PYQ Q1.

Day	Study target	What to write/practise
3	Radio, microwave, infrared	RF propagation and microwave diagrams.
4	Satellite communication	Full satellite diagram and PYQ Q4.
5	Impairments and noise	Attenuation/distortion/noise diagrams and PYQ Q5.
6	Nyquist and Shannon	At least 10 calculations; PYQ Q6/Q7.
7	Timed mock revision	Write Q1–Q8; redraw every diagram from memory.

Accuracy reminders

- Optical fibre works by total internal reflection, not by a simplistic statement that “cladding reflects light.”
- A zero-noise channel is an ideal theoretical model; practical channels always have some noise.
- Use S/N as a linear ratio in Shannon’s formula. Convert from dB before substitution.
- Bandwidth is measured in Hz; data rate is measured in bps; baud means symbols per second.
- A GEO satellite is approximately 35,786 km above the equator and introduces noticeable propagation delay.

Final message: For top marks, write neatly, use headings, include labelled diagrams, box numerical answers, and revise every PYQ in handwriting at least twice.